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PATH-INTEGRAL FORMULATION OF TWO-DIMENSIONAL GAUGE THEORIES WITH MASSLESS FERMIONS

R.E. GAMBOA SARAVÍ and F.A. SCHAPOSNIK

Laboratorio de Física teórica, Departamento de Física U.N.L.P., Comisión de Investigaciones Científicas de Buenos Aires, Argentina

J.E. SOLOMIN

Departamento de Matemática, U.N.L.P., Comisión de Investigaciones Científicas de Buenos Aires, Argentina

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We study $(1+1)$ dimensional gauge theories [with SU(N) and diagonal SU(N) color symmetry] using the functional integral method. We construct an effective lagrangian by performing a change in the fermionic variables, and then investigate relevant phenomena such as the intrinsic Higgs mechanism and color screening.

1. Introduction

There has been recently much interest in the study of field theoretical models in $(1+1)$ dimensions where properties such as free asymptotic behavior and quark confinement (in some cases via color screening) are to be expected. Since one works in 2-dimensional space-time, computations are simpler and, in some cases, the models are exactly soluble.

This is the case of quantum electrodynamics with a massless fermion in 2 dimensions, QED_2, originally solved by Schwinger [1] and studied in detail using bosonization techniques by Lowestein and Swieca [2]. As is well known, in this model the "photon" acquires a mass via an intrinsic Higgs mechanism [3] and the fermion disappears from the physical spectrum. Of course, the hope is that these features, also expected to be present in 2-dimensional non-abelian gauge theories, will survive a dimensional boost.

Much insight has been gained by studying these models using the method of euclidean functional integrals. In particular, Nielsen and Schroer [4] clarified within this context the relation between topology and confinement in QED_2. In their approach to the problem the integration over fermions induced field configurations which can account for the fine structure of the vacuum.

Another approach using functional methods was developed by Roskies, and one of us [5], after Fujikawa's observation [6] on the non-invariance of the path-integral measure under γ_5 transformations. By performing a change of variables which parallels the "operator fit" of Lowestein and Swieca [2], an effective lagrangian was obtained in ref. [5] showing all the known features of the model. This approach has the advantage of maintaining explicitly the (free) fermion degrees of freedom in the generating functional, thus allowing the direct analysis of the fermionic Green functions in a very simple way. Then, properties such as color neutralization, short-distance behavior, etc., can be discussed directly in terms of the effective lagrangian which gives the exact solution of the model (cf. Maiella, ref. [7]). At the same time, the role played by topology is clarified owing to the relation between the jacobian arising from the change of variables and the local version of the Atiyah-Singer theorem.

It is the purpose of this article to extend this treatment to gauge theories in $(1+1)$ dimensions with colored quarks. To this end we describe in sect. 2 the computation of the jacobian factor in the path-integral measure arising from finite chiral transformations in the non-abelian case. The change of variables has been chosen keeping in mind that which decoupled fermions from gauge fields in QED_2; but in the models that we shall consider, several possibilities arise, and one must be careful owing to the non-abelian character of the theory.

In sect. 3 we discuss the effective lagrangian for different models. We first consider as color symmetry group the maximal abelian subgroup (the torus) of $SU(N)$, $SU(N)_D$. For this case we recover the results of Belvedere et al. [8] and Steinhardt [9] (obtained using bosonization techniques). In our path-integral formulation, the intrinsic Higgs mechanism induced by the (massless) fermions is clearly shown and color screening can be discovered directly from very simple computations of Green functions.

Concerning the complete $SU(N)$ color group, no conclusive results have been reported in the literature, and in fact, there are some contradictions in the conclusions obtained by different authors [8–14]. In our approach, additional complicated terms obscured the analysis of the $SU(N)$ effective lagrangian and we were not able to extract the physical spectrum of the theory. We give, however, some indications about the possibility of having similar phenomena such as those encountered in $SU(N)_D$, at the end of sect. 3.

Finally, in sect. 4 we give a brief summary of our results and conclusions.

2. The change of fermionic variables

We start from the gauge-invariant 2-dimensional (euclidean) lagrangian

$$\mathcal{L} = -\bar{\psi}\not{D}\psi + \tfrac{1}{4}F^a_{\mu\nu}F^a_{\mu\nu}, \tag{2.1}$$

where the fermions are taken in the fundamental (N-dimensional) representation of the gauge group SU(N) and

$$A_\mu = A_\mu^a T^a,$$

T^a being the $N^2 - 1$ generators of SU(N) satisfying

$$[T^a, T^b] = if_{abc}T^c, \qquad \mathrm{tr}^c(T^a T^b) = \tfrac{1}{2}\delta^{ab}.$$

The hermitian operator $\not{D}$ is given by

$$\not{D} = \gamma_\mu D_\mu = \gamma_\mu(i\partial_\mu + gA_\mu)$$

and

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu].$$

The hermitian γ-matrices we are using satisfy the following relations:

$$\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}, \qquad \gamma_\mu\gamma_5 = i\varepsilon_{\mu\nu}\gamma_\nu, \qquad (\mu, \nu = 0, 1),$$

where $\gamma_5 = i\gamma_0\gamma_1$ and $\varepsilon_{01} = -\varepsilon_{10} = 1$.

The generating functional of the Green functions is

$$Z[J, \eta, \bar{\eta}] = \int \mathcal{D}A\, \mathcal{D}\psi\, \mathcal{D}\bar{\psi}$$
$$\times \exp\left[-S[A, \psi, \bar{\psi}] + \int \mathrm{d}^2x \left[A_\mu(x)J_\mu(x) + \bar{\eta}(x)\psi(x) + \bar{\psi}(x)\eta(x)\right]\right], \tag{2.2}$$

where

$$S[A, \psi, \bar{\psi}] = \int \mathrm{d}^2x \{\mathcal{L} + \text{gauge-fixing terms}\}.$$

We shall consider a change in the fermionic variables appearing in the generating functional (2.2) defined by means of the γ_5 transformation:

$$\psi(x) = \mathrm{e}^{\gamma_5\phi(x)}\chi(x) = U_5\chi(x),$$
$$\bar{\psi}(x) = \bar{\chi}\mathrm{e}^{\gamma_5\phi(x)}, \tag{2.3}$$

where $\phi(x)$ is an element of the Lie algebra of SU(N) which will be kept arbitrary for the moment.

In terms of the new fields, the lagrangian (2.1) reads

$$\mathcal{L} = -\bar{\chi}\not{D}'\chi + \tfrac{1}{4}F^a_{\mu\nu}F^a_{\mu\nu}, \tag{2.4}$$

where

$$D'_\mu = i\partial_\mu + gA'_\mu, \tag{2.5}$$

A'_μ being shorthand notation for

$$A'_\mu = A_\mu + \frac{1}{2g}\operatorname{tr}\left[\gamma_\mu U_5 \not{\partial}(U_5)\right]. \tag{2.6}$$

Here, tr stands for the trace of the 2×2 γ-matrices, while tr^c is used to denote the trace of the SU(N) matrices; finally we use Tr to denote the direct product $\mathrm{Tr} = \mathrm{tr}\otimes\mathrm{tr}^c$.

It is easy to prove that A'_μ also belongs to the Lie algebra of SU(N); in fact, from eq. (2.6) we get $\operatorname{tr}^c(A'_\mu) = 0$ and also $A'^+_\mu = A'_\mu$.

In the U(1) case (the Schwinger model), by means of a suitable choice of the field $\phi(x)$ which defines the γ_5 transformation, it is possible to decouple the fermionic fields from the bosonic ones, i.e. $A'_\mu \equiv 0$ in eq. (2.5). On the other hand, as pointed out by Fujikawa [6], the path-integral measure is not invariant under the γ_5 transformation (2.3), and it has been proved that [5] it is just the jacobian factor which gives rise to the correct mass term in the path-integral formulation of the model:

$$\frac{e^2}{2\pi}\int A_\mu(x)A_\mu(x)\,\mathrm{d}^2x.$$

The non-triviality of the jacobian, and hence the origin of the intrinsic Higgs mechanism, is related to the axial anomaly [6].

Guided by these results, we shall study the change of variables defined by eq. (2.3) for the SU(N) case. We thus have to express the generating functional (2.2) in terms of the new fields χ and $\bar{\chi}$ and evaluate the corresponding jacobian.

In order to do this, we begin by considering the normalized eigenvectors of the hermitian operator $\not{D}$,

$$\not{D}\varphi_n(x) = \lambda_n\varphi_n(x),$$

and

$$\langle\varphi_n|\varphi_m\rangle = \int \mathrm{d}^2x\,\varphi^+_n(x)\varphi_m(x) = \delta_{nm}. \tag{2.7}$$

The classical fields $\psi(x)$ and $\bar{\psi}(x)$ can be expanded as

$$\psi(x) = \sum_n a_n \varphi_n(x)$$

$$\bar{\psi}(x) = \sum_n \varphi_n(x)^+ \bar{b}_n, \tag{2.8}$$

where a_n and $\bar{b}_n$ are elements of a Grassman algebra [15]. The functional integral measure is then defined by

$$\mathcal{D}A_\mu \mathcal{D}\bar{\psi}\mathcal{D}\psi = \mathcal{D}A_\mu \prod_{n,m} \mathrm{d}\bar{b}_n \mathrm{d}a_m. \tag{2.9}$$

The new fields $\chi(x)$ and $\bar{\chi}(x)$ defined in eq. (2.3) can also be expanded in terms of the φ_n's:

$$\chi(x) = \sum_m a'_m \varphi_m(x),$$

where

$$a'_m = \sum_n c_{m,n} a_n,$$

with

$$c_{m,n} = \langle \varphi_m | U_5^{-1} | \varphi_n \rangle . \tag{2.10}$$

Therefore, the jacobian factor arising from the γ_5 transformation is given by the relation

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi = (\det c_{m,n})^2 \mathcal{D}\bar{\chi}\mathcal{D}\chi. \tag{2.11}$$

(The exponent 2 comes from the fact that the jacobian for $\mathcal{D}\bar{\psi}$ gives rise to an identical factor to the $\mathcal{D}\psi$ one.)

In order to compute the determinant of the c-matrix defined in eq. (2.10), let us consider the one parameter dependent γ_5 transformation

$$\psi(x) = U_5(\alpha)\chi,$$

and

$$\bar{\psi}(x) = \bar{\chi} U_5(\alpha), \tag{2.12}$$

where $U_5(\alpha) = \mathrm{e}^{\gamma_5 \phi(x)\alpha}$ and α is a real parameter to be varied from 0 to 1, allowing us to grow the whole transformation (2.3) by iteration of the infinitesimal one:

$$1+\gamma_5\phi(x)\delta\alpha. \tag{2.13}$$

We should recall that Fujikawa [6] procedure, consists in regularizing, by means of a cut-off M, the trace

$$\sum_n \langle\varphi_n|\gamma_5\phi(x)|\varphi_n\rangle$$

coming from eq. (2.10) in the evaluation of the $(\det c_{n,m})$ when α is infinitesimal; the base $\{\varphi_n\}$ must be identified with the base of eigenvectors of the operator $\not{D}=i\not{\partial}+g\not{A}$ appearing in the lagrangian. In the case of a finite transformation one should note that the operator $\not{D}$ changes step by step when the whole transformation is constructed by iterating the infinitesimal one [eq. (2.13)]. Then, in order to follow Fujikawa's arguments at each stage of the transformation, one must take the eigenvectors which correspond to the covariant derivative appearing in the lagrangian at that stage.

In terms of the new fermionic fields defined in eq. (2.12) the lagrangian (2.1) is

$$\mathcal{L}=-\bar{\chi}\tilde{\not{D}}(\alpha)\chi+\tfrac{1}{4}F^a_{\mu\nu}F^a_{\mu\nu},$$

where the hermitian operator $\tilde{\not{D}}(\alpha)$ is defined as

$$\tilde{\not{D}}(\alpha)=U_5(\alpha)\not{D}U_5(\alpha)=i\not{\partial}+g\tilde{\not{A}}(\alpha), \tag{2.14}$$

with

$$\begin{aligned}\tilde{\not{A}}(\alpha)&=U_5(\alpha)\not{A}U_5(\alpha)+\frac{i}{g}U_5(\alpha)\not{\partial}U_5(\alpha)\\&=\not{A}+\frac{1}{g}U_5(\alpha)\not{D}(U_5(\alpha)).\end{aligned} \tag{2.15}$$

This α-dependent field satisfies

$$\begin{gathered}\frac{\partial\tilde{\not{A}}(\alpha)}{\partial\alpha}=-\gamma_5\tilde{\not{D}}(\alpha)\phi,\\ \tilde{\not{A}}(\alpha)\big|_{\alpha=0}=\not{A},\qquad \tilde{\not{A}}(\alpha)\big|_{\alpha=1}=\not{A}'.\end{gathered} \tag{2.16}$$

We shall consider the eigenvectors $\tilde{\varphi}_n$ of $\tilde{\not{D}}(\alpha)$,

$$\tilde{\not{D}}(\alpha)\tilde{\varphi}_n(x,\alpha)=\tilde{\lambda}_n(\alpha)\tilde{\varphi}_n(x,\alpha),$$

as a complete set of orthonormal fields which we shall take as new bases for the fermionic fields.

Let us now consider the matrix

$$B_{p,n}(\beta;\alpha)=\langle\tilde{\varphi}_p(x,\beta+\alpha)|U_5^{-1}(\alpha)|\tilde{\varphi}_m(x,\beta)\rangle\,, \tag{2.17}$$

which clearly satisfies:

$$B(\beta;\alpha+\delta)=B(\beta+\alpha;\delta)B(\beta;\alpha). \tag{2.18}$$

Taking $\beta=0$ and $\delta=\delta\alpha$ we obtain

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\{\ln\det B(0;\alpha)\}=\frac{\ln\det(B(\alpha;\delta\alpha))}{\delta\alpha}. \tag{2.19}$$

The r.h.s. of this equation can be easily evaluated by taking into account that

$$|\tilde{\varphi}_n(\alpha+\delta\alpha)\rangle\simeq|\tilde{\varphi}_n(\alpha)\rangle+\sum_p b_{np}|\tilde{\varphi}_p(\alpha)\rangle\delta\alpha, \tag{2.20}$$

where $b_{np}=0$ if $n=p$. Then eq. (2.19) becomes

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\{\ln\det B(0;\alpha)\}=-w(\alpha), \tag{2.21}$$

where

$$w(\alpha)=\sum_n\langle\tilde{\varphi}_n(\alpha)|\gamma_5\phi|\tilde{\varphi}_n(\alpha)\rangle\,. \tag{2.22}$$

Now, integrating both sides of eq. (2.21) we obtain:

$$\det\langle\varphi'_p|U_5^{-1}|\varphi_m\rangle=\exp\left(-\int_0^1 w(\alpha)\,\mathrm{d}\alpha\right), \tag{2.23}$$

where we have used the relations

$$B_{nm}(0;0)=\langle\varphi_n|\varphi_m\rangle=\delta_{nm},$$

$$B_{nm}(0;1)=\langle\varphi'_n|U_5^{-1}|\varphi_m\rangle\,.$$

We can write from eq. (2.10)

$$\det c_{m,n}=\det\langle\varphi_m|\varphi'_p\rangle\det\langle\varphi'_p|U_5^{-1}|\varphi_m\rangle\,, \tag{2.24}$$

but it is straightforward to prove from eq. (2.20) following the procedure which led to eq. (2.23) that:

$$\det\langle\varphi_m|\varphi'_p\rangle = 1\,.$$

Therefore, we finally find

$$\mathcal{D}\bar{\psi}\mathcal{D}\psi = \mathcal{D}\bar{\chi}\mathcal{D}\chi\, \mathrm{e}^{-2W}, \tag{2.25}$$

where

$$W = \int_0^1 w(\alpha)\,\mathrm{d}\alpha\,. \tag{2.26}$$

In order to compute the exponent W, we can write from eq. (2.22)

$$w(\alpha) = \int \mathrm{d}^2x\, \mathrm{tr}^{\mathrm{c}}(\phi(x)\omega(x,\alpha)), \tag{2.27}$$

where

$$\omega(x,\alpha)^{ba} = \sum_n \tilde{\varphi}_n^{+a}\gamma_5\tilde{\varphi}_n^b\,. \tag{2.28}$$

This summation is an ill-defined quantity and we evaluate it by using a well-known regularization procedure:

$$\omega(x,\alpha)^{ba} = \lim_{M\to\infty} \sum_n \tilde{\varphi}_n^{+a}(x)\gamma_5\left(\mathrm{e}^{-\tilde{\not{D}}^2/M^2}\right)^{bc}\tilde{\varphi}_n^c(x), \tag{2.29}$$

getting

$$w(\alpha) = \frac{-ig}{8\pi}\int \mathrm{d}^2x\, \mathrm{Tr}\left\{\gamma_5\gamma_\mu\gamma_\nu\phi(x)\tilde{F}_{\mu\nu}(x,\alpha)\right\}, \tag{2.30}$$

where

$$\tilde{F}_{\mu\nu}(x,\alpha) = \partial_\mu\tilde{A}_\nu - \partial_\nu\tilde{A}_\mu - ig\left[\tilde{A}_\mu,\tilde{A}_\nu\right]. \tag{2.31}$$

Making use of the γ properties and the antisymmetry of $\tilde{F}_{\mu\nu}$ we obtain:

$$\begin{aligned} w(\alpha) &= -\frac{ig}{4\pi}\int \mathrm{d}^2x\,\mathrm{Tr}\left\{\gamma_5\phi(\not{\partial}\tilde{\not{A}} - ig\tilde{\not{A}}\tilde{\not{A}})\right\} \\ &= \frac{-g^2}{4\pi}\int \mathrm{d}^2x\left[\frac{1}{2}\frac{\partial}{\partial\alpha}\mathrm{Tr}(\tilde{\not{A}}\tilde{\not{A}}) + \mathrm{Tr}\left(\gamma_5\tilde{\not{A}}\phi\tilde{\not{A}}\right)\right], \end{aligned} \tag{2.32}$$

where we have used eq. (2.16) and neglected a surface term. Therefore, we finally get

$$w(\alpha)=\frac{-g^2}{4\pi}\int \mathrm{d}^2x\left\{\frac{\partial}{\partial\alpha}\operatorname{tr}^{\mathrm{c}}\left(\tilde{A}_\mu\tilde{A}_\mu\right)+i\varepsilon_{\mu\nu}\operatorname{tr}^{\mathrm{c}}\left(\tilde{A}_\mu\left[\tilde{A}_\nu,\phi\right]\right)\right\}, \tag{2.33}$$

since

$$\operatorname{tr}(\gamma_\mu\gamma_\nu)=2\delta_{\mu\nu}, \qquad \operatorname{tr}(\gamma_5\gamma_\mu\gamma_\nu)=2i\varepsilon_{\mu\nu}.$$

Thus, the exponent W appearing in the jacobian of eq. (2.25) is:

$$W=-\frac{g^2}{4\pi}\int \mathrm{d}^2x\left\{\operatorname{tr}^{\mathrm{c}}\left(A'_\mu A'_\mu\right)-\operatorname{tr}^{\mathrm{c}}\left(A_\mu A_\mu\right)+i\varepsilon_{\mu\nu}\int_0^1\operatorname{tr}^{\mathrm{c}}\left(\tilde{A}_\mu\left[\tilde{A}_\nu,\phi\right]\right)\mathrm{d}\alpha\right\}. \tag{2.34}$$

Now, let us show that for the Schwinger model the jacobian contribution [eqs. (2.25) and (2.34)] gives rise to the "photon" mass term, as obtained in ref. [5]. For this abelian case, of course, $[\tilde{A}_\nu,\phi]=0$ and in the Lorentz gauge one can always write:

$$A_\mu=-\frac{1}{g}\varepsilon_{\mu\nu}\partial_\nu\phi,$$

which leads to $A'_\mu=0$, thus decoupling the fermions in the lagrangian. Therefore, from eqs. (2.25) and (2.34) we obtain for the Schwinger model

$$Z=\int\mathcal{D}\bar{\chi}\mathcal{D}\chi\mathcal{D}A_\mu\exp\left[-\int\left(-i\bar{\chi}\partial\!\!\!/\chi+\tfrac{1}{4}F_{\mu\nu}F_{\mu\nu}+\frac{g^2}{2\pi}A_\mu A_\mu\right)\mathrm{d}^2x\right]. \tag{2.35}$$

3. Analysis of the effective lagrangian

As we stated in sect. 1 our aim is to perform a suitable change of variables in the path integral which defines the generating functional of the theory, eq. (2.2), hoping that this change shall (i) decouple fermions from gauge fields in the lagrangian and (ii) give rise to gluon mass terms through the jacobian contribution as a manifestation of the intrinsic Higgs mechanism already encountered in the Schwinger model.

Our guideline will be precisely QED_2, where the solution of Lowestein and Swieca [2] suggested the right change of fermionic variables yielding to an effective lagrangian with a "photon" mass term and decoupled fermions. Of course, the non-commutativity of the $\mathrm{SU}(N)$ algebra makes the task difficult and makes it impossible for us to find a closed solution, except in the case where the whole symmetry group is reduced to the maximal abelian subgroup (the torus) $\mathrm{SU}(N)_{\mathrm{D}}$ of $\mathrm{SU}(N)$ color.

3.1. THE WHOLE GAUGE GROUP IS THE TORUS

We begin by chosing a gauge condition

$$\partial_\mu A^t_\mu = 0, \qquad t \in \text{torus}, \tag{3.1}$$

where t indicates the $(N-1)$ components of the gauge field when the color group is the torus of SU(N).

Condition (3.1) allows us to take ϕ defining the γ_5 transformation satisfying

$$A^t_\mu = -\frac{1}{g}\varepsilon_{\mu\nu}\partial_\nu\phi^t. \tag{3.2}$$

Then, eq. (2.15) reduces to

$$\tilde{A\!\!\!/}(\alpha) = (1-\alpha)A\!\!\!/, \tag{3.3}$$

yielding $A'_\mu = 0$. Of course, all commutators vanish in this case and then the exponent W becomes

$$W = \frac{g^2}{8\pi}\int d^2x\, A^t_\mu A^t_\mu. \tag{3.4}$$

In order to write the generating functional (2.2) in terms of the transformed fields $\bar{\chi}$ and χ we define an effective lagrangian containing also the contribution of the jacobian factor [see eq. (2.25)]. Hence,

$$S_{\text{eff}}[A,\bar{\chi},\chi] = S[A,\bar{\chi},\chi] + 2W[A] \equiv \int d^2x\, \mathcal{L}_{\text{eff}}(A,\bar{\chi},\chi), \tag{3.5}$$

where

$$\mathcal{L}_{\text{eff}} = -\bar{\chi}i\partial\!\!\!/\chi + \tfrac{1}{4}F^t_{\mu\nu}F^t_{\mu\nu} + \frac{g^2}{4\pi}A^t_\mu A^t_\mu. \tag{3.6}$$

This lagrangian corresponds to N free massless fermion fields and $N-1$ gauge fields with mass $g/\sqrt{2\pi}$.

We shall now use the effective lagrangian (3.6) in the generating functional in order to compute fermionic Green functions. Of course one must replace $\bar{\psi}$ and ψ in the fermion source terms in terms of the new fields $\bar{\chi}$ and χ and the relation between ϕ and A_μ [eq. (3.2)] must be taken into account.

Then, it is easy to compute exact fermion Green functions showing that the behavior found by Casher, Kogut and Susskind [16] for the Schwinger model extends to SU$(N)_{\text{D}}$ color symmetry.

As an example, the two-point Green function is given by the expression

$$\begin{aligned} G^{ab}_{\alpha\beta}(x) &\equiv \langle \psi^a_\alpha(x)\bar{\psi}^b_\beta(0)\rangle \\ &= G^0_{\alpha\beta}(x)\delta_{ab}\exp\left[\frac{N-1}{2N}\left\{K_0\left(\frac{gx}{\sqrt{2\pi}}\right)+\ln\left(\frac{gx}{\sqrt{2\pi}}\right)+\gamma\right\}\right], \end{aligned} \tag{3.7}$$

where G^0 is the euclidean Green function of the free Dirac equation

$$G^0(x-y)=\text{const}\times\frac{\gamma_\mu(x_\mu-y_\mu)}{|x-y|^2}, \tag{3.8}$$

and the argument in the exponential corresponds to the difference of the Feynman propagators of a massive and a massless scalar fields.

$$\Delta_F(m^2,x)-\Delta_F(0,x)=-\left(K_0\left(\frac{gx}{\sqrt{2\pi}}\right)+\ln\left(\frac{gx}{\sqrt{2\pi}}\right)+\gamma\right). \tag{3.9}$$

The presence of $\Delta_F(0,x)$ is a manifestation, in the path-integral formulation, of the zero-mass (gauge) excitations which appear in the operator solution.

Now, when $x^2 \to 0$, $\Delta_F(m^2,x)-\Delta_F(0,x)\to 0$ and the short-distance behavior of the exact fermion propagator is characteristic of free fermions.

On the other hand, for large x^2,

$$G^{ab}_{\alpha\beta}(x)\sim \text{const}\times G^0_{\alpha\beta}(x)\delta_{ab}x^{(N-1)/2N}. \tag{3.10}$$

Then, for large times, $G(x)$ tends to infinity relative to $G^0(x)$. As discussed in detail in ref. [16] this cannot be interpreted as if the probability to find a fermion at a later time became infinite. Indeed the result is gauge dependent and was obtained in the Lorentz gauge. Passing to the more physical Coulomb gauge, the probability of finding a single fermion at any later time vanishes (see ref. [16]).

This completes the picture of the theory and agrees with the results found in refs. [8] using a generalization of the Lowestein-Swieca operator solution: no asymptotic fermions exist and the intermediate spectrum is exhausted by $(N-1)$ bosons of mass $g/\sqrt{2\pi}$, although the short-distance behavior is identical to that of free fermions in two dimensions.

All matrix elements of composite fermion fields relevant for the computation of structure functions can be easily obtained following the procedure described above.

3.2. THE GAUGE GROUP IS SU(N)

The first question that arises when one attacks the complete SU(N) color symmetry case concerns the possibility of finding a relation between A_μ and ϕ which decouples all the fermions in the lagrangian (2.1). From eq. (2.15) it is evident that decoupling occurs if ϕ satisfies the relation:

$$A = -\frac{i}{g}(\not{\partial} U_5)U_5^{-1}. \tag{3.11}$$

It was not possible for us to show rigorously that for any A_μ there always exists a ϕ such that relation (3.11) holds.

We then leave this possibility for a moment and pass to a second question that helps in the understanding of the model: which is the most convenient gauge to work with? A candidate which simplifies considerably the computations is the Lorentz gauge

$$\partial_\mu A_\mu^t = 0, \qquad t \in \text{torus}, \tag{3.12}$$

for the components of A_μ belonging to the torus, while for those outside it, one takes

$$F_{\mu\nu}^a = 0, \qquad a \notin \text{torus}. \tag{3.13}$$

These conditions belong to the class of hybrid gauges defined by Baluni [10] and used in refs. [9, 10]. Eq. (3.13) expresses the fact that it is always possible to choose $F_{\mu\nu}$ pointing in the torus direction. Furthermore, one can perform gauge transformations which maintain $F_{\mu\nu}$ in the torus and at the same time lead to A_μ's satisfying eq. (3.12).

We want to note at this point that it is not possible to make all the A_μ's equal to zero outside the torus, except if the symmetry is broken, *manu militari*, down to $SU(N)_D$ as done in ref. [14]. This invalidates the obvious choice $A_\mu^a = 0, a \notin$ torus instead of eq. (3.13) which would solve exactly the theory in its path-integral formulation.

Condition (3.12) allows us to take ϕ with no components outside the torus and satisfying

$$A_\mu^t = -\frac{1}{g}\varepsilon_{\mu\nu}\partial_\nu \phi^t, \qquad t \in \text{torus}. \tag{3.14}$$

Note that ϕ is related only with the torus components of the gauge field, in contrast to relation (3.11).

With this, eq. (2.15) reduces to

$$\tilde{\not{A}}(\alpha) = \not{A}^t(1-\alpha) + U_5(\alpha)\not{A}^\perp U_5(\alpha), \tag{3.15}$$

and then

$$\not{A}' = U_5 \not{A}^\perp U_5 = \not{A}^\perp + U_5\gamma_\mu\left[A_\mu^\perp, U_5\right], \tag{3.16}$$

where $A_\mu^\perp$ are the components of A_μ orthogonal to the torus.

One can evaluate explicitly the α-integral appearing in W [$W = -\frac{1}{2}\ln(\text{jacobian})$, eq. (2.34)] as follows:

$$i\varepsilon_{\mu\nu}\int_0^1 \mathrm{tr}^c\big(\tilde{A}_\mu[\tilde{A}_\nu,\phi]\big)\,\mathrm{d}\alpha = -\int_0^1 \mathrm{Tr}\big(\gamma_5 \tilde{\not{A}}\tilde{\not{A}}\big)\,\mathrm{d}\alpha$$

$$= -\mathrm{tr}^c\big(A^t_\mu A^t_\mu\big)\int_0^1 (1-\alpha)\,\mathrm{d}\alpha - \tfrac{1}{2}\int_0^1 \frac{\partial}{\partial\alpha}\,\mathrm{tr}^c\big(\tilde{A}_\mu\tilde{A}_\mu\big)\,\mathrm{d}\alpha$$

$$= \tfrac{1}{2}\,\mathrm{tr}^c\big(A^\perp_\mu A^\perp_\mu\big) - \tfrac{1}{2}\,\mathrm{tr}^c\big(A'_\mu A'_\mu\big).$$

Thus, eq. (2.34) can be written

$$W = \frac{g^2}{4\pi}\int \mathrm{d}^2x\Big\{\mathrm{tr}^c\big(A^t_\mu A^t_\mu\big) + \tfrac{1}{2}\,\mathrm{tr}^c\big(A^\perp_\mu A^\perp_\mu\big) - \tfrac{1}{2}\,\mathrm{tr}^c\big(A'_\mu A'_\mu\big)\Big\}. \qquad (3.17)$$

But, using eq. (3.16), we finally obtain

$$W = \frac{g^2}{4\pi}\int \mathrm{d}^2x\Big\{\tfrac{1}{2}A^t_\mu A^t_\mu + \tfrac{1}{4}\,\mathrm{Tr}\big(\gamma_\mu\gamma_\nu\big[\mathrm{e}^{2\gamma_5\phi}, A^\perp_\nu\big]A^\perp_\mu\,\mathrm{e}^{-2\gamma_5\phi}\big)\Big\}. \qquad (3.18)$$

Then, the effective lagrangian is

$$\mathcal{L}_{\mathrm{eff}} = -\bar{\chi}\big(i\not{\partial} + g\gamma_\mu \mathrm{e}^{-\gamma_5\phi}A^\perp_\mu \mathrm{e}^{\gamma_5\phi}\big)\chi + \tfrac{1}{4}F^t_{\mu\nu}F^t_{\mu\nu}$$
$$+ \frac{g^2}{4\pi}A^t_\mu A^t_\mu + \frac{g^2}{8\pi}\,\mathrm{Tr}\big[\gamma_\mu\gamma_\nu\big[\mathrm{e}^{2\gamma_5\phi}, A^\perp_\nu\big]A^\perp_\mu\,\mathrm{e}^{-2\gamma_5\phi}\big]. \qquad (3.19)$$

It is worth noting that the term $g\mathrm{e}^{-\gamma_5\phi}A^\perp_\mu\mathrm{e}^{\gamma_5\phi}$, the only one which remains coupled to the fermionic current in the effective lagrangian, has no components in the torus. In this sense one could say that the new fermionic fields $\bar{\chi}$ and χ only interact with an effective field carrying no torus color degrees of freedom. Since $F^\perp_{\mu\nu} = 0, A^\perp_\mu$ appears in $\mathcal{L}_{\mathrm{eff}}$ only through self-interactions or interactions with A^t_μ and χ.

We see that the effective lagrangian (3.19) is rather complicated. Even for the simplest SU(2) case it takes the form

$$\mathcal{L}_{\mathrm{eff}} = -\bar{\chi}\big(i\not{\partial} + g\gamma_\mu A^a_\mu T^a \mathrm{ch}(\varphi) + g\gamma_\nu\varepsilon_{\mu\nu}\varepsilon_{ab}A^a_\mu T^b \mathrm{sh}(\varphi)\big)\chi$$
$$+ \tfrac{1}{4}F^3_{\mu\nu}F^3_{\mu\nu} + \frac{g^2}{4\pi}A^3_\mu A^3_\mu - \frac{g^2}{4\pi}A^a_\mu A^a_\mu \mathrm{sh}^2(\varphi)$$
$$+ \frac{g^2}{8\pi}A^a_\mu A^b_\nu \mathrm{sh}(2\varphi)\varepsilon_{\mu\nu}\varepsilon_{ab}, \qquad (3.20)$$

with a mass term for A^3_μ, but interactions expressed in terms of $\phi = \varphi T^3$ and $A^{1,2}_\mu$

obscure any direct interpretation. We then return to the first possibility mentioned at the beginning of this section, eq. (3.11),

$$\not{A} = -\frac{i}{g}(\not{\partial} U_5)U_5^{-1},$$

which would lead to $A' = 0$, hence completely decoupling the fermion fields in $\mathcal{L}_{\mathrm{eff}}$ [in contrast with the choice (3.14) which led to the effective lagrangian (3.19)]. As we stated before, we were not able to show rigorously that all A_μ can be written in this form. Moreover, even if this were possible it is not clear if the gauge is automatically fixed by this relation as it occurs for relation (3.14) in the abelian case.

Leaving aside these serious objections one can proceed to compute the jacobian factor. The resulting effective lagrangian takes, in this case, the form

$$\mathcal{L}_{\mathrm{eff}} = -\bar{\chi} i \not{\partial} \chi + \tfrac{1}{4} F^a_{\mu\nu} F^a_{\mu\nu} + \frac{g^2}{4\pi} A^a_\mu A^a_\mu$$
$$+ \frac{g^2}{4\pi} \varepsilon_{\mu\nu} f_{abc} \phi^c \int_0^1 \tilde{A}^a_\mu(\alpha) \tilde{A}^b_\mu(\alpha)\,\mathrm{d}\alpha . \qquad (3.21)$$

The presence of the fourth term in eq. (2.37) obscures again the interpretation of the effective lagrangian. A superficial analysis indicates that the $N^2 - 1$ A_μ's have gotten a mass $g/\sqrt{2\pi}$. Even if this result was obtained under restrictive assumptions, it points in a direction already signaled in the literature [8]: the intrinsic Higgs mechanism encountered for $\mathrm{SU}(N)_{\mathrm{D}}$ color symmetry should also occur for $\mathrm{SU}(N)$ as well as all the implicances of this on color screening.

4. Conclusions

We have studied in this article two-dimensional gauge theories with massless fermions using the functional integral method. We believe that this approach considerably simplifies the analysis and provides useful insights into the intrinsic Higgs mechanism induced by the massless fermions and the color screening characteristic of this type of theories.

The clue to our approach lies in Fujikawa's [6] observation on the non-invariance of the functional integral measure under chiral transformations. By a suitable change of variables, precisely involving a γ_5 transformation, we obtained an effective lagrangian which allows, in some cases, a direct analysis of the theory.

Using this method, a kind of path-integral version of the Lowestein-Swieca operator solution for QED_2, we studied two different models. The first one corresponds to diagonal $\mathrm{SU}(N)$ color symmetry and we were able to reproduce in a very simple way the results obtained by Belvedere et al. [8] and Steinhardt [9]: the $(N-1)$ gauge fields get a mass $m_{\mathrm{A}} (m_{\mathrm{A}} = g/\sqrt{2\pi})$ and decouple from the fermions.

These behave as if they were free at short distances, but color screening forbids the existence of real asymptotic fermions. These properties were derived from the generating functional constructed with our effective lagrangian. It is important to note that in this approach the (free) fermion degrees of freedom appear explicitly in the generating functional in contrast with the customary path-integral formulation where one integrates completely the fermion variables [4]. This is the reason why fermion Green functions and, in general, any matrix element of composite quantum fields relevant for the computation of structure functions, can be constructed in a very economical way by using our method.

Concerning the case of SU(N) color symmetry, additional complicated terms made the analysis of our effective lagrangian difficult. We were then not able to decide which of the possible (and sometimes contradictory) scenarios proposed for this model actually holds [8–14]. However, we believe that there are some indications about the possibility of decoupling the fermion fields. Hence, one could expect the intrinsic Higgs mechanism encountered for SU$(N)_{\mathrm{D}}$ to extend to SU(N) as suggested in ref. [8], but we have no satisfactory proof of this assertion.

We want to end this article by noting that, as already stressed by Fujikawa [6], there is a deep relation between the jacobian of chiral transformations and the Atiyah-Singer index theorem. Since it was through the jacobian that phenomena like the intrinsic Higgs mechanism and screening became manifest in our approach, we believe that it should be worth studying this kind of path-integral change of variables in QCD_4 where topological properties make possible the non-triviality of the jacobian.

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